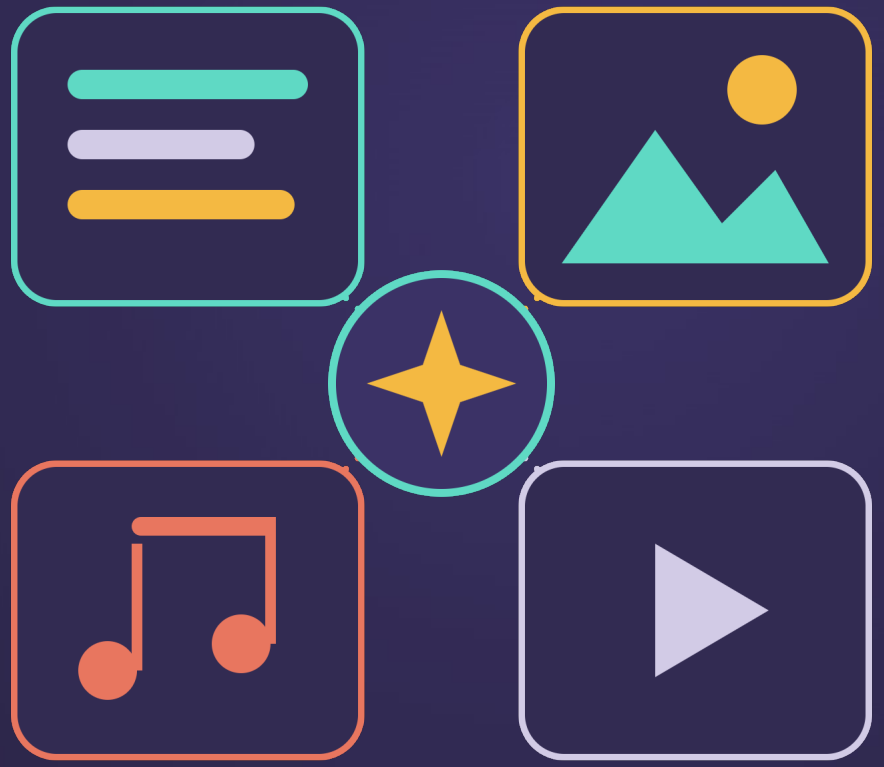
WEEK 5 · SATURDAY

What's new in AI

Beyond the chatbot: AI that sees, hears and reasons.

AI for Beginners · Lesson 5, Part A (Current trends)



TODAY'S PLAN

What AI can do now, and how to use it well

LAST WEEK IN ONE LINE

A chatbot predicts the **next token**, using **attention**.

Today: what else that idea has unlocked.

1. **Beyond text**
   How AI makes images, video and sound, and how it thinks.
2. **Using AI well**
   Prompting, and giving it your own documents.
3. **Small, open, local**
   Right-sizing the model, and AI in Africa.
4. **Responsible use**
   New AI threats, real hacks, and the new laws.

WHERE WE ARE

From Week 1 to today

1956
"Artificial intelligence" gets its name

1997
A computer beats the chess world champion

2012
Deep learning wins at recognising images

2017
The transformer and attention arrive

2022
ChatGPT brings chatbots to everyone

Now
Sees, hears, reasons and acts

You have met every dot on this line in Weeks 1 to 4.

Today and tomorrow are about the last dot.

PART 1 · GENERATIVE AI

Describe it, and it makes it

Text
Answers, essays, translations

Images
Pictures from a description

Audio
Voices, music, sound effects

Video
Short clips from a prompt

Code
Programs and fixes

A PROMPT

"A busy market in Addis Ababa at sunset, watercolour style"

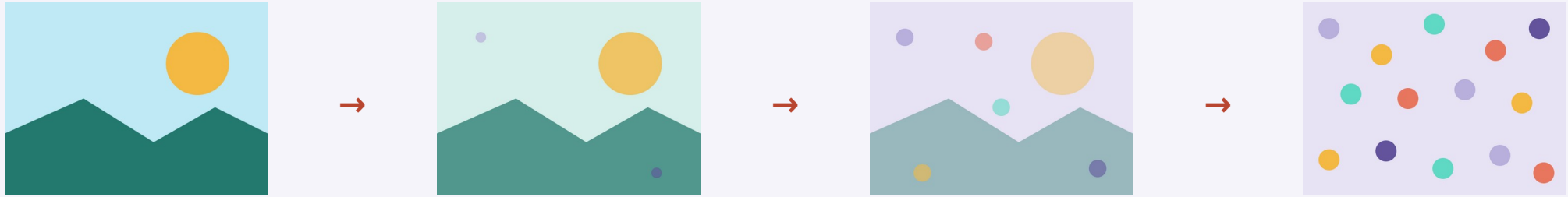
Same recipe every time: learn patterns from huge data, then create something new that fits them.

Looks real is not real. Generated images and clips can mislead.

PART 1 · HOW IMAGE TOOLS LEARN

Training: spoil a photo, then fix it

Step A Add noise on purpose



Real photo → A little noise → More noise → Pure static

Step B Practise undoing it

1 Show a noisy picture
+ its caption

→

2 Guess the noise
"Which specks were added?"

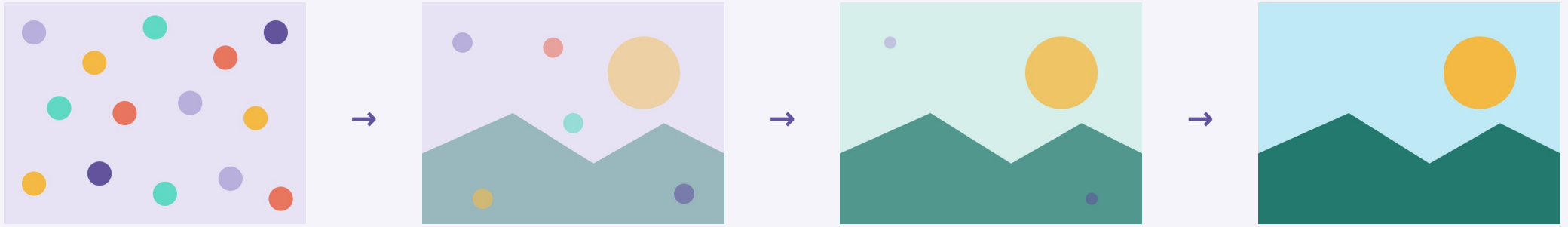
→

3 Check and adjust
We know the right answer

> **No labels needed:** we made the mess, so we know the answer.

Generating: start from static, clean it up

Prompt: **"a sun over green hills"**



1. Random static → **2. Remove a little** → **3. Repeat ~30 times** → **4. Finished image**

Your words steer every cleaning step.

New static, new picture every time.

PART 1 · BEYOND DIFFUSION

Many ways to make pictures, video and sound

Diffusion and flow

Clean static, step by step

USED FOR

Most images and video

Token by token

Predict the next piece, like a chatbot

USED FOR

Chatbot images, voices, music

GAN

Forger vs detective, in one step

USED FOR

Filters, upscaling, voices

Variational Autoencoders

A VAE compresses an image, then rebuilds it

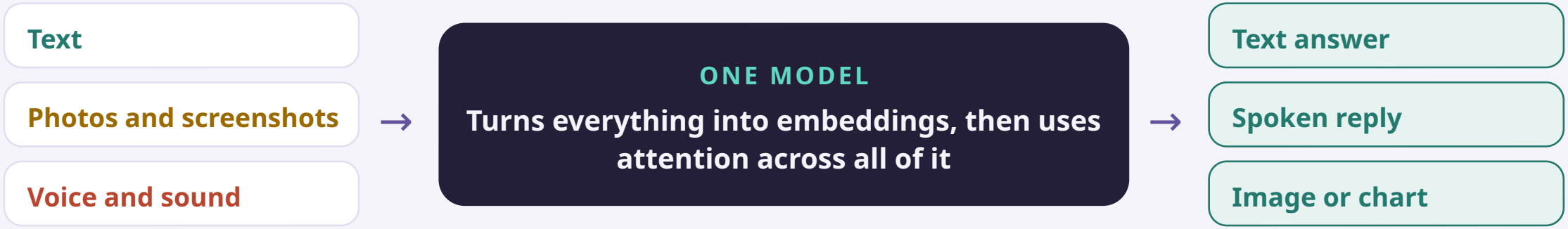
USED FOR

A helper inside diffusion tools

Same recipe: learn patterns from huge data, then make something new.

PART 1 · MULTIMODAL

One model, many senses



Farmer

Photo of a spotted maize leaf: "What is wrong?"

Student

Photo of a hard maths problem: "Explain step by step."

Everyone

Speak a question, hear the answer. No typing.

Still check: a model can misread a photo just as it can invent a fact.

PART 1 · REASONING MODELS

Think first, then answer

Puzzle: a bus leaves at 8:40 and the trip takes 2 hours 35 minutes. When does it arrive?

Answers straight away

"It arrives at 11:05."

Fast, but wrong

Works it out first

8:40 + 2 hours = 10:40

10:40 + 35 minutes = 11:15

"It arrives at 11:15."

Slower, and right

How: the model writes its working as extra tokens before the final answer.

Trade-off: more time and cost. Best for maths, logic, code and planning.

PART 2 · PROMPTING

Ask clearly, get better answers

Weak prompt

"Tell me about photosynthesis."

Strong prompt

"You are a biology teacher. Explain photosynthesis to a 14-year-old in 5 bullet points, using a farm example. End with 2 quiz questions."

1. Role

Who should it act as?

2. Task

What exactly to do?

3. Context

Who is it for? What do you have?

4. Format

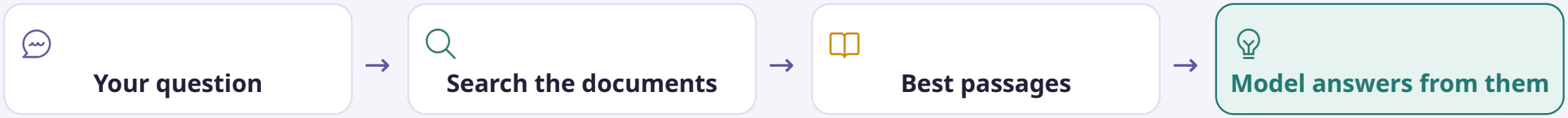
List, table, length, tone?

Bonus moves: show an example of what you want, and ask it to ask you questions first.

Iterate: the first answer is a draft. Reply "shorter", "simpler", "more examples".

PART 2 · GROUND IT IN FACTS

Let it look things up first



Example: a school's help bot

Question: "When is the fee deadline?"

Found: page 3 of the school handbook

Answer: "Fees are due on 15 March (handbook, page 3)."

Helps with: made-up facts, old knowledge, and private documents.

Not magic: if the search finds the wrong page, the answer is wrong.

PART 3 · RIGHT SIZE, RIGHT PLACE

Bigger is not always better

Big cloud models

Most capable. Need internet. You pay per use and send your data away.

Small models

Fast and cheap. Can run on a laptop or phone, even offline. Good at one job.

Open-weight models

Download them, run them privately, adapt them. Check the licence.

Which one fits?

A hard, one-off task needing top quality	→ **Big cloud model**
A clinic app in a village with weak internet	→ **Small on-device model**
Student records that must not leave the school	→ **Open model, run locally**

Rule of thumb: pick the smallest model that does the job well.

PART 3 · AI AT WORK

Where AI helps people today

Learning

Explain, quiz and give feedback on your own pace.

Health

Help doctors read scans and notes. Humans decide.

Farming

Spot crop disease from a photo. Plan around weather.

Software

Suggest code, find bugs, write tests.

Languages

Translate, caption and read aloud.

Creating

Design, music, video and storytelling.

Pattern: AI suggests. A person checks and decides.

Your turn: where would AI help in your family or town?

PART 3 · CLOSE TO HOME

AI for Africa: gaps and chances

THE GAP

2,000+

languages are spoken in Africa. Most have very little online text or recorded speech.

Less data means weaker AI.

Community projects

Groups such as Masakhane build language data and tools together.

Local data

Collect speech and text with consent, with fair credit and benefit.

Local problems

Voice tools for farmers, health workers and learners in their own language.

Your edge: you speak these languages and know these problems. That is what the data and the tools are missing.

PART 4 · RISKS

What can go wrong

Fakes and scams

Cloned voices, fake photos, fake videos used to trick people.

Misinformation

False stories made quickly, and shared even faster.

Unfairness

Biased data gives biased decisions, for example in hiring or loans.

Privacy

Personal data typed into tools or collected without consent.

Over-reliance

Letting AI do the thinking, so your own skills fade.

Changing jobs

Some tasks fade, new ones appear. Skills must keep up.

Two questions for any AI project: who could be harmed? Who is left out?

PART 4 · A NEW LEVEL OF THREAT

AI makes attacks cheaper, faster and bigger

	BEFORE AI	WITH AI
Scam messages	Slow, full of mistakes	**Perfect, in any language**
Faking a person	Needed an actor	**Seconds of voice is enough**
Finding weak spots	Rare experts, weeks	**Thousands of bugs found**
Running an attack	A skilled team	**One person + an AI agent**

PART 4 · CASE FILE, OCTOBER 2026

The $6.3 billion Binance heist

THE THEFT

$6.3B

taken from Binance on 29 October 2026

FBI blames North Korea. North Korea denies it.

JUNE	**Fake AI coding tool installed**
SEPTEMBER	**Fake job interview, booby-trapped test**
20 OCTOBER	**Internal AI tool used to take over laptops**
29 OCTOBER	**Binance's own systems sign the withdrawals**

Lesson: the blockchain was never broken. People and their laptops were.

PART 4 · NOT JUST ONE CASE

AI in real attacks, and in defence

2024 · HONG KONG

Deepfake boss on a video call

$25 million sent to scammers

2025 · WORLDWIDE

An AI agent did the hacking

80 to 90% of the work, ~30 targets

2026 · MEXICO

Jailbroken chatbot helps one hacker

150 GB of government data stolen

2026 · DEFENCE

An AI too risky to release

Found thousands of bugs, for defenders

The pattern: AI does not break in by magic. It makes tricks faster and cheaper.

PART 4 · RULES OF THE ROAD

Who sets the rules for AI?

Laws

Some, like the EU AI Act, sort AI by risk. Higher risk means stricter rules.

Countries and regions

National AI policies, and the African Union's continental AI strategy.

Companies

Usage rules, safety tests, and limits on what their models will do.

Schools and workplaces

Rules on when AI help is allowed, and when it must be declared.

A risk ladder (the idea behind many laws)

Not allowed | **Strict checks** | **Be transparent** | **Mostly free**

Still changing: rules differ by country and update often. Look up the latest for your own.

PART 4 · LAWS RACING TO CATCH UP

What's happening with AI laws right now

2024
Ethiopia and the AU: AI strategies

2025
EU bans the worst uses

Jan 2026
Korea: label AI content

Feb 2026
Kenya bill: jail for harmful deepfakes

Jun 2026
EU delays rules, bans 'nudify' apps

Oct 2026
Denmark: your face is yours

The big debate

"Protect people first"
Harms spread fast. Rules should come early.

"Don't slow innovation"
Heavy rules are costly. Wait and see.

Discuss: copy the EU, write our own rules, or wait?

RECAP

Four things to remember

Beyond text

AI makes images, video and sound, using diffusion and more.

Think and look up

Use thinking mode for hard problems. Documents ground the answer.

Right size

Small, open and local models often beat the giant one for the job.

Use it wisely

Verify everything, expect smarter scams, know the rules.

Quiz: the Week 5 trends quiz is on Google Forms. It tests judgement, not memory.

Tomorrow: AI agents. Models that plan, use tools and take action.

BEFORE TOMORROW

Try it yourself

1. Write a **weak** and a **strong** prompt for the same task: a study plan for your next exam.
2. A **village clinic** has weak internet and sensitive records. Which model setup, and why?
3. Your "cousin" sends a **voice note** asking for money, fast. What 2 things do you do?